Baselines: architectures and results

EF estimation on EchoNet-Dynamic; every baseline evaluated with the same protocol as the proposed sync model

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Table B1 EF accuracy of B1-B8 (validation and test)

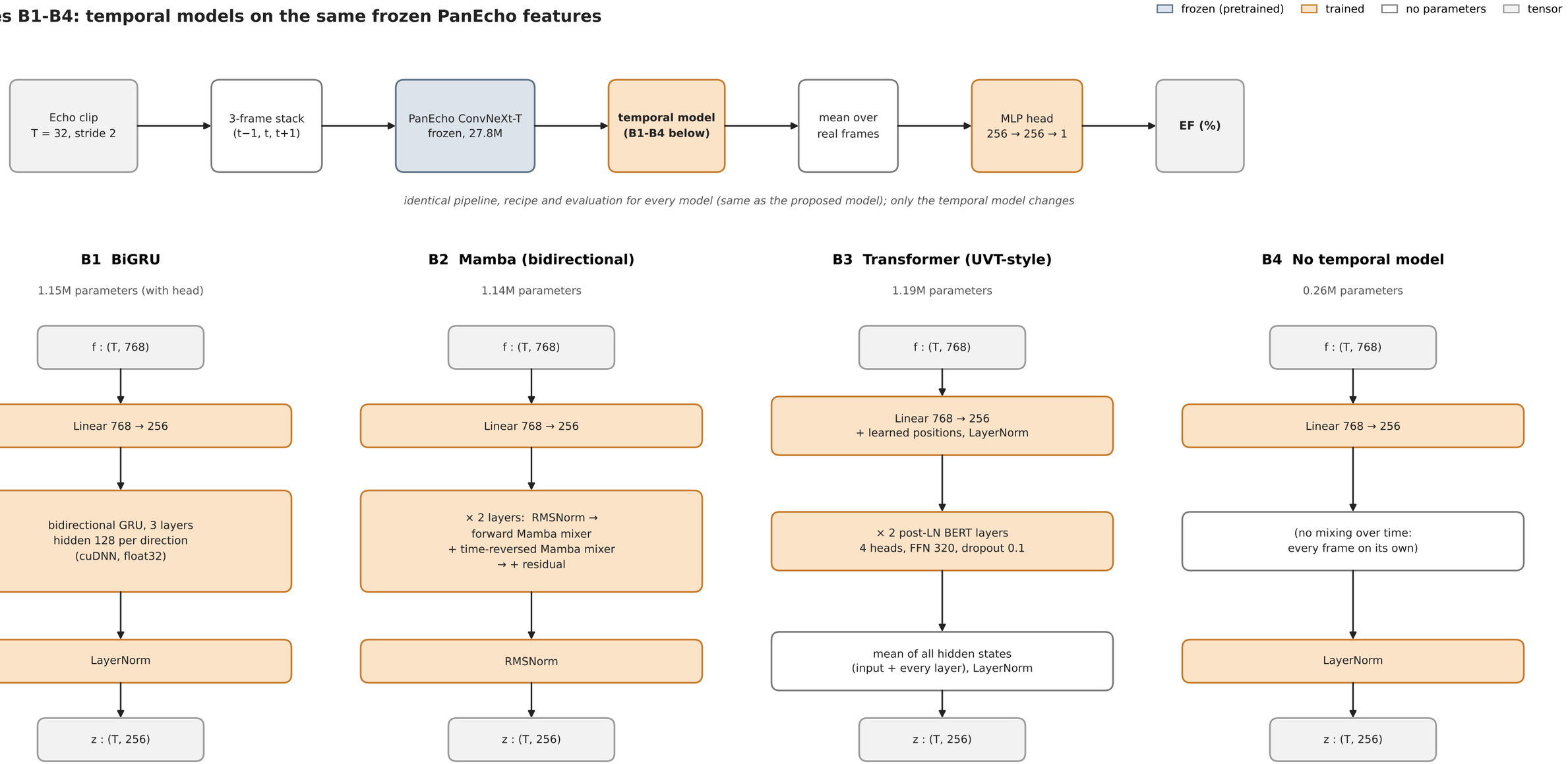
Table B2 Temporal behaviour and cost of B1-B4 (validation and test)

Table B3 End-to-end baselines B9-B11 (validation and test)

The proposed model (sync, fixed frequency) appears as the last column of Tables B1-B2 for reference only; its architecture is in ../architecture.

Data: EchoNet-Dynamic, Stanford AIMI (doi:10.71718/yqp5-y078), research use only. PanEcho (Holste et al., JAMA 2025), weights CC BY-NC-SA 4.0.

Baselines B1-B4: temporal models on the same frozen PanEcho features



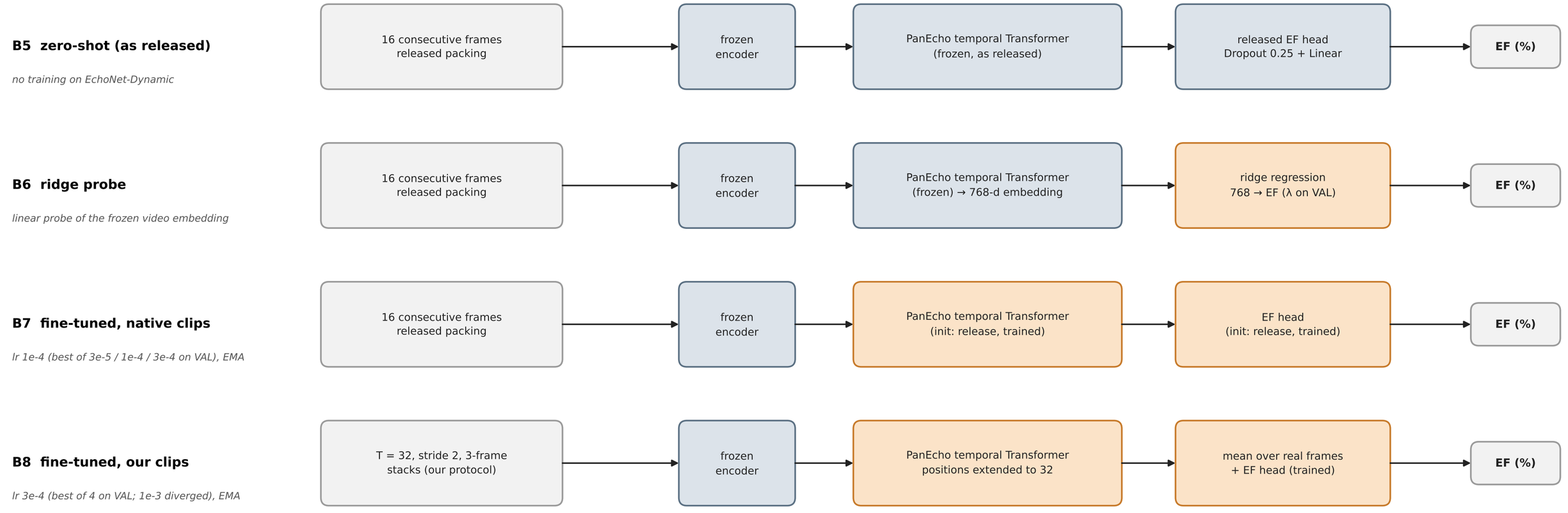
Mamba mixer (B2, one direction): in_proj 256 → 2 × 256 (expand 1) → causal depthwise conv (k = 4) + SiLU → x_proj → Δ (rank 16), B, C (state N = 16) → selective scan (S6, A = −exp(A_log), S4D-real init, + D skip) → y ⊙ SiLU(z) → out_proj 256 → 256. The backward mixer sees the clip reversed in time; the two are summed.

Training as the proposed model: MSE on standardised EF, AdamW lr 1e-3, cosine, batch 16, 40 epochs, weight EMA 0.999; 3 seeds; evaluation on 4 clips per video.

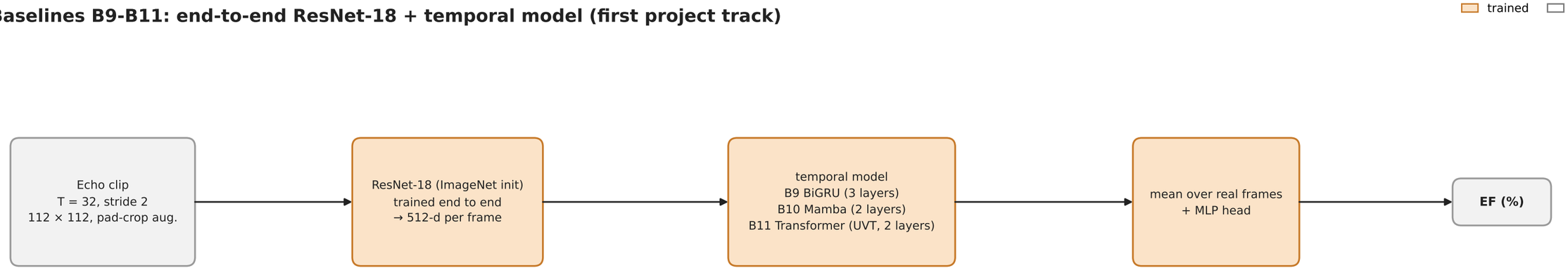
Baselines B5-B8: PanEcho's own video model (Holste et al., JAMA 2025)

frozen (pretrained) | trained | no parameters | tensor

shared, frozen: PanEcho ConvNeXt-T image encoder (27.8M) — per-frame 768-d features, cached once. Transformer: 4 post-norm layers, d 768, 8 heads, FFN 768, ReLU, sinusoidal positions, mean over time (14.2M parameters).



Baselines B9-B11: end-to-end ResNet-18 + temporal model (first project track)



Same temporal blocks as B1-B3 with 512-d input. AdamW lr 1e-4, cosine, batch 16, 25 epochs, seed 0, no EMA (this track predates it). Superseded by the frozen-PanEcho track (B1-B8), which is better for every model.

Table B1. EF accuracy of the baselines — validation set

Every model uses the same frozen PanEcho ConvNeXt-T frame encoder.
proposed: sync temporal block variants | baselines on the same frozen features (+ weight EMA) | PanEcho baselines

	BiGRU	Mamba	Transformer (UVT)	no temporal model	PanEcho TF our clips	PanEcho TF 16 frames	PanEcho ridge probe	PanEcho zero-shot	sync fixed (final) [reference]
trained parameters	1.15M	1.14M	1.19M	0.26M	14.19M	14.19M	769	0	1.15M
seeds	3	3	3	3	3	3	1	1	3
MAE (EF points) ↓	4.056	3.998	4.236	4.573	4.139	4.609	5.030	5.382	**3.963**
RMSE ↓	5.424	5.293	5.658	6.059	5.476	6.114	6.772	7.341	**5.253**
R^2 ↑	0.806	0.815	0.789	0.758	0.802	0.753	0.697	0.644	**0.818**
Pearson r ↑	0.900	0.903	0.890	0.871	0.896	0.869	0.838	0.807	**0.905**
CCC ↑	0.890	0.896	0.889	0.863	0.889	0.853	0.810	0.784	**0.899**
|bias| ↓	0.739	0.070	**0.056**	0.316	0.175	0.121	0.143	1.079	0.188
within 5 points (%) ↑	70.1	70.3	68.0	65.9	69.2	65.0	60.8	59.8	**71.1**
within 10 points (%) ↑	93.0	**93.8**	92.1	90.2	93.1	89.9	87.2	85.5	93.7
AUROC, EF < 40% ↑	0.983	0.983	0.981	0.978	0.982	0.976	0.969	0.959	**0.984**
AUROC, EF < 50% ↑	0.959	0.961	0.957	0.945	**0.961**	0.947	0.936	0.921	0.960
MAE difference vs BiGRU ↓	reference	-0.057	+0.180	+0.517	+0.084	+0.553	+0.974	+1.326	**-0.093**
95% CI of the difference	—	[-0.135, +0.019]	[+0.066, +0.290]	[+0.370, +0.663]	[-0.012, +0.176]	[+0.405, +0.705]	[+0.777, +1.174]	[+1.114, +1.544]	[-0.164, -0.020]
p vs BiGRU (paired bootstrap)	—	0.139	0.002	< 0.001	0.086	< 0.001	< 0.001	< 0.001	0.011

Validation set (1,288 videos), EF averaged over 4 evenly spaced clips per video, metrics averaged over seeds. Paired statistics over videos with seed-averaged absolute errors (10,000 bootstrap resamples).
Our clips: T = 32 frames at stride 2 (63-frame window); PanEcho's native clips: 16 consecutive frames. All models except the zero-shot one are trained on EchoNet-Dynamic TRAIN only; design choices made on validation.
Trained parameters: temporal model + head (the frozen PanEcho image encoder, 27.8M, is shared and excluded); ridge = 768 weights + bias. Bold = best in the row.

Table B1. EF accuracy of the baselines — test set

Every model uses the same frozen PanEcho ConvNeXt-T frame encoder.
proposed: sync temporal block variants | baselines on the same frozen features (+ weight EMA) | PanEcho baselines

	BiGRU	Mamba	Transformer (UVT)	no temporal model	PanEcho TF our clips	PanEcho TF 16 frames	PanEcho ridge probe	PanEcho zero-shot	sync fixed (final) [reference]
trained parameters	1.15M	1.14M	1.19M	0.26M	14.19M	14.19M	769	0	1.15M
seeds	3	3	3	3	3	3	1	1	3
MAE (EF points) ↓	4.105	**4.020**	pending	pending	pending	pending	pending	pending	4.043
RMSE ↓	5.444	5.313	pending	pending	pending	pending	pending	pending	**5.302**
R^2 ↑	0.802	0.811	pending	pending	pending	pending	pending	pending	**0.812**
Pearson r ↑	0.900	0.901	pending	pending	pending	pending	pending	pending	**0.902**
CCC ↑	0.886	0.893	pending	pending	pending	pending	pending	pending	**0.894**
|bias| ↓	0.995	**0.287**	pending	pending	pending	pending	pending	pending	0.444
within 5 points (%) ↑	69.4	**70.7**	pending	pending	pending	pending	pending	pending	70.2
within 10 points (%) ↑	93.0	**93.4**	pending	pending	pending	pending	pending	pending	93.2
AUROC, EF < 40% ↑	0.981	0.982	pending	pending	pending	pending	pending	pending	**0.983**
AUROC, EF < 50% ↑	0.967	**0.967**	pending	pending	pending	pending	pending	pending	0.966
MAE difference vs BiGRU ↓	reference	**-0.085**	pending	pending	pending	pending	pending	pending	-0.062
95% CI of the difference	—	[-0.159, -0.007]	pending	pending	pending	pending	pending	pending	[-0.132, +0.011]
p vs BiGRU (paired bootstrap)	—	0.032	pending	pending	pending	pending	pending	pending	0.093

Test set (1,277 videos), EF averaged over 4 evenly spaced clips per video, metrics averaged over seeds. Paired statistics over videos with seed-averaged absolute errors (10,000 bootstrap resamples).
Our clips: T = 32 frames at stride 2 (63-frame window); PanEcho's native clips: 16 consecutive frames. All models except the zero-shot one are trained on EchoNet-Dynamic TRAIN only; design choices made on validation.
Trained parameters: temporal model + head (the frozen PanEcho image encoder, 27.8M, is shared and excluded); ridge = 768 weights + bias. Bold = best in the row.

Table B2. Temporal behaviour and cost of the baselines — validation set

Baselines on the same frozen PanEcho 3-frame-stack features (B1-B4, B8).

proposed: sync temporal block variants | baselines on the same frozen features (+ weight EMA) | PanEcho baselines

	BiGRU	Mamba	Transformer (UVT)	no temporal model	PanEcho TF our clips	sync fixed (final) [reference]
MAE at T = 8 ↓	**5.861**	8.520	11.241	6.626	—	5.998
MAE at T = 16 ↓	4.504	4.615	5.036	4.797	—	**4.288**
MAE at T = 32 (trained) ↓	4.056	3.998	4.236	4.573	—	**3.963**
MAE at T = 64 ↓	4.024	4.009	4.339	4.495	—	**3.989**
MAE at T = 128 ↓	**4.064**	4.091	4.428	4.485	—	4.112
MAE, heart rate < 60 bpm (n = 321) ↓	**3.636**	3.708	3.915	4.290	—	3.648
MAE, heart rate 60–100 bpm (n = 884) ↓	4.130	4.040	4.277	4.558	—	**4.013**
MAE, heart rate > 100 bpm (n = 83) ↓	4.897	4.676	5.032	5.819	—	**4.656**
EF SD across a video's 4 clips ↓	1.345	1.407	1.537	**1.330**	—	1.476
feature smoothness, cos(z_t, z_t+1) (descriptive)	0.926	0.812	0.968	0.898	—	0.890
periodicity, autocorrelation peak (descriptive)	0.144	0.192	0.229	0.237	—	0.185
ED frame probe error (frames) ↓	2.093	2.239	**1.968**	2.649	—	2.125
ES frame probe error (frames) ↓	2.315	2.129	**1.793**	2.391	—	2.128
MAE with frames shuffled (higher = uses order)	15.468	15.455	4.245	4.573	—	14.835
forward GFLOP per clip ↓	0.069	0.068	0.069	**0.013**	0.919	0.072
latency, ms per clip (batch 16) ↓	0.040	1.101	0.026	**0.007**	0.334	0.082
training step, ms (batch 16) ↓	3.6	56.6	5.6	**1.3**	16.4	10.6

Validation set, mean of 3 seeds, 4 clips per video. Models trained once at T = 32 (stride 2) and evaluated at every T. Heart rate: label-free whole-video estimate from the frame features (scripts/hr_reference.py).
Probes: ridge regression from the frozen block outputs to the ED/ES frame (fitted on TRAIN, lambda on validation); error in sampled frames. Smoothness and periodicity are descriptive, not accuracy.
Cost: temporal model + head only (the shared frozen encoder is excluded), RTX 3050 laptop GPU, bf16, CUDA graphs; '—' = not measured or not applicable (PanEcho TF has no per-frame evaluation outputs).

Table B2. Temporal behaviour and cost of the baselines — test set

Baselines on the same frozen PanEcho 3-frame-stack features (B1-B4, B8).

proposed: sync temporal block variants | baselines on the same frozen features (+ weight EMA) | PanEcho baselines

	BiGRU	Mamba	Transformer (UVT)	no temporal model	PanEcho TF our clips	sync fixed (final) [reference]
MAE at T = 8 ↓	**5.795**	8.470	pending	pending	—	5.917
MAE at T = 16 ↓	4.675	4.595	pending	pending	—	**4.381**
MAE at T = 32 (trained) ↓	4.105	**4.020**	pending	pending	—	4.043
MAE at T = 64 ↓	**4.043**	4.085	pending	pending	—	4.064
MAE at T = 128 ↓	**4.049**	4.198	pending	pending	—	4.140
MAE, heart rate < 60 bpm (n = 296) ↓	4.030	**3.874**	pending	pending	—	3.926
MAE, heart rate 60–100 bpm (n = 904) ↓	4.065	**4.018**	pending	pending	—	4.043
MAE, heart rate > 100 bpm (n = 77) ↓	4.865	4.599	pending	pending	—	**4.505**
EF SD across a video's 4 clips ↓	**1.341**	1.390	pending	pending	—	1.466
feature smoothness, cos(z_t, z_t+1) (descriptive)	0.925	0.811	pending	pending	—	0.890
periodicity, autocorrelation peak (descriptive)	0.144	0.193	pending	pending	—	0.185
ED frame probe error (frames) ↓	2.211	2.377	pending	pending	—	**2.119**
ES frame probe error (frames) ↓	2.414	**2.255**	pending	pending	—	2.269
MAE with frames shuffled (higher = uses order)	15.539	15.360	pending	pending	—	14.699
forward GFLOP per clip ↓	0.069	0.068	0.069	**0.013**	0.919	0.072
latency, ms per clip (batch 16) ↓	0.040	1.101	0.026	**0.007**	0.334	0.082
training step, ms (batch 16) ↓	3.6	56.6	5.6	**1.3**	16.4	10.6

Test set, mean of 3 seeds, 4 clips per video. Models trained once at T = 32 (stride 2) and evaluated at every T. Heart rate: label-free whole-video estimate from the frame features (scripts/hr_reference.py).
Probes: ridge regression from the frozen block outputs to the ED/ES frame (fitted on TRAIN, lambda on validation); error in sampled frames. Smoothness and periodicity are descriptive, not accuracy.
Cost: temporal model + head only (the shared frozen encoder is excluded), RTX 3050 laptop GPU, bf16, CUDA graphs; '—' = not measured or not applicable (PanEcho TF has no per-frame evaluation outputs).

Table B3. End-to-end ResNet-18 baselines (first project track, seed 0)

4 clips per video on TEST (T = 32 at stride 2 unless stated); VAL = best single-clip MAE of the training log. Lower is better; bold = best.

model	parameters (encoder + temporal + head)	VAL MAE (best single clip)	TEST MAE	TEST RMSE	TEST R^2	TEST MAE T = 8	TEST MAE T = 16	TEST MAE T = 64	TEST MAE T = 128
B9 BiGRU	12.26M	**4.474**	**4.569**	**6.133**	**0.748**	7.392	5.015	4.618	4.662
B10 Mamba	12.25M	4.595	4.636	6.271	0.737	**6.795**	**4.969**	**4.601**	**4.630**
B11 Transformer (UVT)	12.30M	4.649	4.663	6.332	0.732	10.035	5.291	4.700	4.731

Trained before the frozen-PanEcho track with lr 1e-4, 25 epochs, pad-crop augmentation and no weight EMA; these numbers are not comparable with Tables B1-B2, where every model shares the frozen PanEcho encoder.